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Intrusion Detection Systems in IoT Based on Machine Learning: A state of the art

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Abstract

Nowadays, connected devices are progressively integrating our lifestyle, known as the Internet of Things (IoT). There are several domains where these IoT can be found, including industry, smart cities, energy, healthcare, smart homes, and so on. These devices bring with them a host of security threats, due to their many weaknesses such as limited resources and the fact that it is often out of date. This is why researchers have progressively turned to innovative techniques to face these threats, such as Intrusion Detection Systems (IDS). In addition, IDS can be enhanced through machine learning (ML), which improves detection accuracy, adapts to new and evolving threats, identifies unknown threats, reduces false positives, and more. In this study, we conduct a review of the recent articles focusing on the integration of IDS and ML. Each method is evaluated based on attributes such as feature selection method, algorithms used, detection and false positive rates, security threats, and so on. This involves presenting a critical analysis of each approach. Furthermore, we present the most prevalent datasets containing deliberate attacks on test IoT networks. In addition, we cover the anomaly detection-based IDS method in depth, focusing on papers that use ML. Subsequently, we engage in a discussion by comparing all methods and identifying open re-search questions. As a result, summarized tables are provided to highlight each model.

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1. Introduction

In a world where various technology domains are evolving at an exponential pace, such as sensors, wireless communications with ultra-powerful speeds like 5G, and soon to be 6G (promising speeds up to 100 times faster than 5G), as well as high-speed internet and embedded systems, the introduction of smart devices into our daily lives has become possible. The communication between these smart devices over the Internet has given rise to a new term: the Internet of Things (IoT). According to a study published on Statista, an online portal specializing in statistics, the number of connected devices could reach 30.9 billion by 2025 [1]. Additionally, the IoT market is expected to continue growing between 2024 and 2028, generating an estimated revenue of \$145.4 billion, representing a 54.75% increase [2]. There are several domains where these devices can be found, including industry, smart cities, energy, healthcare, smart homes, and agriculture, among others [3]. However, this digital transformation has also introduced new security threats. Attacks on critical infrastructure such as defense and nuclear facilities are on the rise [4]. These attacks have an impact not only on the country itself but also on our environment.

In addition, the existence of multiple standards and various communication channels makes it difficult for traditional security methods to adapt to the IoT world.

To face these issues, several security solutions are already available to enhance IoT security, cryptographic techniques such as homomorphic encryption [5], and authentication [6]. In addition, the techniques used to maintain trust between the various components of the IoT paradigm [7]. Despite the multitude of security measures, they remain insufficient against network-targeted attacks. This raises the question of whether there is a method to add a new layer of defense capable of detecting and responding to security threats in a proactive and timely manner. The solution to this question lies in the field of Intrusion Detection Systems (IDS). IDS involves the continuous monitoring of network traffic and computer events to identify unexpected occurrences. However, IDSs need to be adapted to IoT devices. In addition, the existing traditional IDS models can be bypassed and sometimes fail to detect new attacks. Some researchers have shifted towards anomaly-based IDS category and machine learning (ML) methods due to their ability to learn from data to detect these new attacks.

This work provides a review of the recent articles focusing on the integration of IDS and ML. The primary contributions outlined in this working paper include:

- This work presents an overview of the IDS ecosystem by defining the IDS types and existing detection methods.
- We explore the key elements for model analysis by presenting the most prevalent datasets containing deliberate attacks on test IoT networks, as well as discussing various feature reduction techniques and evaluation criteria.
- This work conducts a detailed study of the most recent articles (2017-2023) presenting intrusion detection methods based on artificial intelligence. We cover the anomaly detection-based IDS method in depth, focusing on papers that use ML.
- We engage in a discussion by comparing all methods, identifying open issues in IDS development for IoT, and indicating future research directions.

The remainder of this paper is structured as follows: In Section 2, we present an overview of an IDS. Before exploring ML techniques for IDS, Section 3 outlines key elements: common datasets, preprocessing methods, and evaluation metrics. Section 4 is devoted to the latest research aimed at enhancing ML-based IDS systems in IoT. We conclude with a synthesis of the work presented in this paper in Section 5 and outline our future work in Section 6.

2. Intrusion detection systems (IDS) ecosystem: an overview

Intrusion detection involves the continuous monitoring of network traffic and computer events to identify unexpected occurrences. When performed using a software application, it is denoted as IDS. IDS is a type of network security designed to identify and acknowledge potential threats prior to services being disrupted, unauthorized access being granted, or data being compromised. Two categories of IDS can be distinguished: Host IDS (HIDS) [8] and Network IDS (NIDS) [9]. First, The **HIDS** analyses the activity of a system by monitoring all changes to files, including system files, as well as memory usage. This approach is executed on all available hosts in

a network, monitoring each host and determining whether suspicious activity is occurring. Second, **NIDS** are strategically located devices in networks. They passively examine the traffic as it traverses across the devices where they are installed. Network IDS can come in the form of either hardware or software systems. It can interface with network technologies such, as Ethernet, FDDI and more depending on the manufacturer. NIDS devices are equipped with two network interfaces. One interface is solely dedicated to monitoring network communications in promiscuous mode, while the second interface serves control and reporting purposes. The difference between the two types is that HIDS focuses on the security of individual hosts by analysing internal system data, while NIDS monitors the overall network traffic to detect suspicious activities across multiple devices.

In addition, we can distinguish several IDS detection strategies, **Signature Detection System** [10] is a detection technique that relies on a signature database. Essentially, this technique continuously analyses network traffic, searching for any matches with existing patterns of known attacks, either generated by a NIDS or a HIDS. The major limitation of this technique is its inability to catch new attacks, which means that the signature database must always be kept up to date, potentially allowing attackers to infiltrate the system as it relies on regular expressions.

The **anomaly IDS** [11] works by creating a reference point of what constitutes normal network behaviour. then continuously monitors network traffic to detect any deviations from this baseline indicating malicious activities. When an anomaly is identified the IDPS sends alerts to take actions to block attacks. Based on this construction, it can identify unknown attack patterns that don't have specific signatures. It is also adaptive and dynamic as it can learn from evolving network behaviour and adjust the baseline accordingly. Nevertheless, there are challenges associated with anomaly-based detection. Establishing and maintaining a baseline can be difficult in complex and diverse network environments. Additionally, false negatives can occur where malicious traffic blends, with data or imitates behaviour leading to missed attacks.

The **hybrid-based** IDS leverages the advantages of both methodologies to achieve superior results.

3. Key Elements for ML Model Analysis

Before delving into different ML techniques for IDS, it's crucial to understand three key elements: commonly used datasets, dataset preprocessing methods, and evaluation metrics. This section focuses on dataset preprocessing, highlighting feature selection and extraction, and explores metrics for evaluating model performance.

3.1. Common Datasets

To properly assess the robustness of an ML algorithm, it is important to target the right dataset that combines both quantity and quality of data, as well as the assets we aim to predict or identify (see the paper [12] for more details). We present below some of the most used datasets in the IoT domain:

- **Canadian Institute for Cybersecurity IDS(CICIDS-2017):** Developed by the CIC, this dataset is freely available for research, featuring 15 classes, 80 features, and 7 attack categories, including DoS, PortScan, and Brute Force. It encompasses traffic data from diverse network services like FTP, HTTP, SSH, etc., catering to various research needs. Several noteworthy papers use this dataset. [13,14]
- **UNSW-NB15:** This dataset, created by the University of New South Wales (UNSW) in Australia, includes nine attack categories like Fuzzers, DoS, and Reconnaissance. With 49 features extracted from network traffic, it's ideal for ML tasks. Its substantial size, with 2.5 million records, enables robust NIDS model training and evaluation. It has been utilized in studies [15, 16, 17].
- **NSL-KDD:** This dataset is an enhanced iteration of the original KDD Cup 99 dataset, designed to address limitations of the original version. With 150,000 samples and 41 attributes, it retains the same attack classes. The attacks are categorized into Denial of Service (DoS), User-to-Root (U2R), Remote-to-Local (R2L), and Probe. It has been employed in various studies [13, 14, 18].
- **IoT-23:** It was developed by the NSL (Network Security Lab) at the University of Colorado Boulder. It comprises a total of 20 features. It contains 20 malware captures (including Mirai, Torii, Gagfyt, Kenjiro, ...) performed on IoT devices, with three benign IoT device traffic captures. It was used in [19, 20]

- **Kyoto:** It comprises daily records of real network traffic data spanning from 2006 to 2009. It encompasses a total of 24 distinct features. Specifically, it includes 50,033,015 sessions of normal traffic, 43,043,225 sessions of identified attacks, and 425,719 sessions categorized as unknown attacks.
- **CSE-CIC-IDS-2018:** Developed by the CIC, this dataset is freely available for research purposes. It comprises 80 features and includes attacks such as Brute force, Portscan, Botnet, DoS, and DDoS.
- **AFDA:** Created by the University of New South Wales, this dataset focuses primarily on Zero-day attacks, Stealth attacks, and Webshell attacks.

3.2. Preprocessing phase: Feature selection (FS)

Feature Selection (FS) is a preprocessing step aiming to identify relevant features while discarding irrelevant ones, crucial for managing computational complexity in datasets. FS aims to create a feature subset that effectively represents the problem. Common methods include ML-based, filter, and wrapper methods. Wrapper methods evaluate feature performance using the learning algorithm to select optimal features, while filter methods choose features without relying on specific ML algorithms, making them efficient for removing duplicated, correlated, and redundant features, though they don't address multicollinearity.

In literature, several studies employ FS to achieve favourable results in accuracy and false positive rates. In the paper [21], the authors select 23 features from the NSL-KDD dataset that vary based on values, meaning these values change in an abnormal situation. Subsequently, they use Random Forest (RF) to calculate the importance of these selected features. The preprocessing is based on the wrapper method, which returns the top 5 most important features from the dataset used. However, no guarantee that unselected features are not required for a specific dataset. Moreover, the authors in [22] introduce a method for identifying wireless network threats that combines deep learning with a wrapper-based approach to reduce feature space. The wrapper method was able to select 26 features of the 154 available attributes. Furthermore, the authors of [23] present a tree-seed algorithm (TSA) aimed at feature extraction. TSA aims to reduce data dimensionality by removing redundant features, ultimately enhancing the performance of the ML classifier in terms of accuracy. This FS method was successful in selecting 6 features from 41, achieving an accuracy of 87.34%. All results showed that the utilisation of FS methods improves the accuracy of the proposed models.

3.3. Evaluation metrics

To assess the robustness of a method in intrusion detection in IoT, several metrics are necessary. The possible results of attack prediction are outlined as follows:

- True positive (TP): It denotes attack classes that were predicted correctly.
- False positive (FP): Identifies normal classes mistakenly categorized as attacks.
- False negative (FN): Denotes attack categories mistakenly identified as normal.
- True negative (TN): Refers to the accurately predicted normal classes.

Using all the parameters presented above, we can determine the Accuracy (A). it denotes the total percentage of correctly identified instances in the test dataset in comparison to the total number of instances, defined as

$$A = \frac{TP + TN}{TP + FN + FP + TN} \quad (1)$$

These metrics are used to evaluate the model. First, the Recall (R) assess the model's capability to correctly predict TPs by computing the proportion of TPs identified among all actual positives. It is defined as

$$R = \frac{TP}{TP + FN} \quad (2)$$

Second, Precision (P) can be determined by calculating the count of TPs divided by the total positives, which includes both false and TPs. It reflects the model's predictive quality, defined as

$$P = \frac{TP}{TP + FP} \quad (3)$$

In addition, we can use the F1-score to assess both precisions and recalls simultaneously, defined as

$$F1-score = 2 * \frac{P * R}{P + R} \quad (4)$$

The F1-score ranges from **0 to 1**, where 1 means perfect precision and recall, and 0 means the worst.

4. Next-Gen NIDS for IoT: ML-based approach

ML-based approaches analyse real-time traffic and provide very satisfactory results compared to other existing solutions. The aim is to learn from the normal usage of the device by analysing patterns. We dedicate this to present three types of approaches: Supervised, Unsupervised, and Deep Learning Methods.

4.1. Most ML algorithms used in IDS

This section defines the two types of most used ML methods. These are, firstly, supervised ML methods and, secondly, unsupervised ML methods.

Concerning supervised ML methods, there are the Artificial Neural Networks (ANN) [14] and Support Vector Machines (SVM) [18]. ANN are a computational model inspired by the intricate structure and function of biological neural networks in the human brain. These networks consist of interconnected nodes organized into layers. Each node receives input data, processes them through activation functions, and transmits outputs to other nodes in the network. Despite their powerful capabilities, methods based on ANNs face significant challenges, including complexity, the need for vast amounts of training data, and extended processing times. Additionally, ANNs can struggle with overfitting, making it difficult to generalize well on unseen data. To address some of these limitations, SVM offer a robust alternative. SVM primary objective is to identify the optimal hyperplane separating different feature space classes. By maximizing the margin between classes, SVM demonstrates robustness against noise and excels in handling high-dimensional data.

Regarding unsupervised ML methods, Deep Neural Networks (DNN) and Convolutional Neural Networks (CNN) are advanced types of neural networks that have shown significant promise in the field of intrusion detection. CNN [16] is an advanced version of ANN primarily used to extract features from grid-like data structures, such as images. CNNs are composed of several layers, including an input layer, convolutional layers, pooling layers, and fully connected layers. The convolutional layers apply filters to the input data to extract significant features, the pooling layers downsample the data to reduce computational complexity, and the fully connected layers make the final predictions. In contrast, DNNs [36] differ from CNNs in a few keyways. Traditional neural networks typically have one or two hidden layers, making them relatively shallow. DNNs, on the other hand, have multiple hidden layers, allowing them to model more complex patterns in the data. This depth enables DNNs to automatically learn high-level features from raw input, making them particularly effective for complex tasks such as intrusion detection.

4.2. Supervised Learning Methods

This category of ML uses data that has already been classified to predict new data by identifying patterns. Several methods have been proposed in the literature, such as ANN [24, 25], SVM [15, 18, 26], Linear Regression (LR) [18], RF [13], Autoencoder (AE) [27] and Naive Bayes (NB) [14].

One of the interesting papers that uses ANN [24] is a computational model inspired by the structure and function of biological neural networks in the human brain. The authors consist of interconnected nodes or neurons organized into layers. Each neuron receives input signals, processes them through activation functions, and passes outputs to other neurons in the network. However, the major issues with the use of ANN are their complexity and the requirement for large amounts of data for training, as well as their long processing times. The following discussion focuses on recent research that uses ANN as a classifier for IDS in IoT. Another approach is presented in [25], where the authors propose the use of ANNIDS to detect two specific attacks in RPL-based networks: the DODAG Information Solicitation (DIS) attack and the Version attack. The ANNIDS model has a five-layer Multi-Layer Perceptron (MLP), including three hidden layers. The data model is trained on 80% of the packets (59,104),

including 1,047 attacks and 14,776 benign packets. The remaining 20% is used for testing, including 260 attack packets. The experiment shows that the TP Rate, P, R, and F1-score values were at their highest levels. However, the author failed to furnish details regarding the dataset employed, and the features selection method. It is unclear whether the dataset used in the study was a custom dataset or one of the well-known datasets commonly used in similar research.

Concerning the approaches based on SVM, the paper [28] focuses on attacks that disrupt IoT network traffic intensity, mainly DoS attacks. The proposed SVM model relies solely on packet arrival to the node. The filter method was used for FS, using three features: the packet arrival rate's minimum, maximum, and median values. When the packet arrival rate surpasses the threshold value, it is classified as an intrusion. The dataset used for training was generated using MATLAB 2018b based on a Poisson distribution due to the absence of the required data attribute, namely, the packet arrival rate, in the datasets available online. The authors show a comparison to other proposed models GA-SVM [29], A-IDS [30], and WFS-IDS [31]. The experiment yields favourable outcomes in accuracy (98.35%), FP lower than 2% and training/testing time that takes 16.3592 seconds. The proposed system appears to be efficient in identifying DoS attacks, but its applicability to other IoT attacks remains a question. One major drawback of the proposed approach is its inability to detect attacks that do not significantly alter traffic intensity. In addition, the authors in [18] developed algorithms to detect threats to smart irrigation in smart farms. Three different algorithms were developed based on three different classification algorithms: LR, SVM and RF. The goal was to test algorithm efficiency and robustness against IoT attacks, crucial for precise watering in varying plant needs. The authors used the NSLKDD dataset and component analysis (PCA) approach for FS. 20% of the data (25,192 records) was allocated for training, while the remaining 80% (100,781 records) was reserved for testing. The comparison of the three algorithms showed a significant superiority of the SVM classifier, achieving 98% accuracy compared to the other algorithms, which did not exceed 78%. Moreover, the proposed systems were only tested on one dataset, prompting concerns about their effectiveness and adaptability to different datasets.

Table 1. Summary of ML methods.

Category	Year	Reference	Approach	Attacks	Dataset	Accuracy	FS	FP
Supervised	2019	Anitha, al [24]	ANN	DIS attack and Version Attack	-	100%	No info	0%
Supervised	2019	Hanif, al. [25]	ANN	Types, encompassing fuzzers, shellcode, worms,	UNSW-NB 15	84%	-	8%
Supervised	2019	Jan, al.[28]	SVM	DoS attack	Custom dataset	98.35%	Filter	<2%
Supervised	2022	Raghuvanshi, al. [18]	SVM, LR, RF	Irrigation system attack	NSL-KDD	98% 78%	PCA	-
Supervised	2021	Keserwani, al. [13]	GWO, PSO, RF	DoS, Probe, R2L, U2R	KDDCup99, NSL-KDD, CICIDS-2017	99.66%	GWO-PSO	-
Unsupervised	2017	Al-Yaseen, al. [32]	SVM, K-Means, ELM	DoS, U2R and R2L attacks	KDD Cup 1999	95.17%	K-Means	1.87%
Unsupervised	2018	Nömm, al. [26]	IF, LOF, SVM	DoS	BASHLITE and Mirai	More than 97%	LCC	-
Unsupervised	2021	Odu-muyiwa, al.[27]	AE, RBM, K-Means, EM	SYN-Flood, UDP-Lag	CICDDOS2019, Mirai, BASHLITE	99% 50% 80%	minmax	-
Unsupervised	2022	Shitharth, al.[14]	ADC, DBSCANPPGO, LNB	DoS, Probe, R2L, U2R	NSL-KDD, CICIDS-2017, Bot-IoT	93.89	PPGO	4.7%
Unsupervised	2021	Liu, al.[15]	OCSVM	types,encompassing fuzzers, shellcode, worms, ...	UNSWNB15	86.68%	PSO, LightGBM	10.62%

4.3. Unsupervised Learning Methods

Unsupervised learning is an approach to ML in which an algorithm is trained on a set of data without predefined labels or responses. To detect Distributed Denial of Service (DDoS) attacks, the authors [27] develop multiple models for detecting SYN-Flood and UDP-Lag attacks focusing on the transport layer. To choose the best unsupervised algorithm, meaning one with higher precision and a very low FP rate, four different methods were employed: Autoencoder (AE), Restricted Boltzmann Machine (RBM), K-Means, and Expectation-Maximization (EM). To train their models, the authors used CICDDOS2019, Mirai, and BASHLITE datasets, and the minmax method for preprocessing. Results favoured the AE algorithm, with over 99% accuracy on Mirai and BASHLITE datasets, while other algorithms ranged from 50% to 80% accuracy. However, several questions arise, such as the percentage of data used to train their methods and the FP rate. Additionally, doubts arise regarding the effectiveness of the minmax preprocessing method on all the algorithms. In addition, the paper [32] proposed a system known as a Hybrid SVM and Extreme Learning Machine (ELM) at Multiple Levels, incorporating a modified K-means approach for intrusion detection. This system combines the advantages of both algorithms to enhance its performance. The authors developed a customized version of K-means, which reduced the size of the dataset by creating five new smaller training datasets that represent the main dataset. These training datasets were then utilized to train the classifiers SVM combined with ELM. The system was trained using the KDD Cup dataset. The approaches decrease the training duration and minimize the FP rate while enhancing the precision in detecting both known and unknown attacks on a specific dataset. However, the authors did not use other datasets to confirm the robustness of their model, and the dataset used in this study is quite old, dating back to 1999, and may suffer from various issues [33].

Moreover, the paper [26] demonstrates the possibility of achieving high-precision unsupervised methods with a reduced set of features. The authors developed a botnet detection system. They opted for three approaches with lower computational complexity (LCC) to select important features. The first approach identifies only features with high entropy, while the second selects features with high variance. The last one utilizes Howkins' statistics. Subsequently, the chosen data and features are evaluated using the following algorithms: isolation forest (IF), local outlier factor (LOF), and SVM. To assess their model, they combined two different datasets: BASHLITE and Mirai. Results show the first approach, based solely on high entropy features, achieves high precision and accuracy with only 5 features. However, the precision score diminishes if the number of features is increased or decreased, raising questions about determining the appropriate number of features to choose.

4.4. Deep Learning Methods

Deep learning [34] is a subfield of ML that uses DNNs to model complex relationships between input data and output predictions [35]. Various DL methods, such as DNN [36] and CNN [16], enable efficient handling of diverse intrusion types and enhance detection accuracy. A compelling approach is introduced in [37]. The method employs two modules: Spider Monkey Optimization (SMO) for feature selection and Stacked Deep Polynomial Networks (SDPN) for attack classification. Trained on the NSL-KDD dataset, the framework outperforms DT, GBT, and KNN with 99% accuracy, precision, recall, and F-score. However, lacking validation on other datasets, and training on 85% of the data, its real-world reliability is questionable.

In addition, [38] employs a feedforward neural network (FNN) for binary and multiclass classification, focusing on DDoS, DoS, Reconnaissance, and Information Theft attacks. Detection time is approximately 5 minutes for a single attack type and 20 minutes for multiple attacks. Trained on the Bot IoT dataset, the system achieves 98% accuracy across various attacks and 88% accuracy for data exfiltration due to its resemblance to reconnaissance.

Table 2. Summary of Deep Learning Methods.

Year	Reference	Approach	Attacks	Dataset	Accuracy	FS	FP
2022	Otoum, al. [37]	S-DPN	DoS, DDoS, R2L, Probe	NSLKDD	99.02%	SMO	-
2019	Ge, al. [38]	FNN	DoS, DDoS, Reconnaissance, Information theft	Bot IoT	98%	-	<2%
2019	Thamilarasu, al. [36]	DBN, DNN	Sinkhole, DDoS, blackhole portunistic service, wormhole	Custom dataset	99%	-	-
2022	Siddharthan, al. [39]	SVM, GB, KNN, NB, RF	SYN Flood	SENMQTT-SET	99%	LR	1.3%
2022	Sayed, al. [16]	MyCNN, IoTCNN	Backdoor, DoS, Fuzzers, Reconnaissance, shellcode, worms, exploits,	NF-UNSW-NB15-v2	99%	-	<2%
2023	Chen, al. [17]	DT, WE	Attacks in (HeIoT) networks	KDD CUP 99, NSL-KDD, UNSW-NB15	more than 98.5%	-	-

5. Discussion

Regarding the datasets used, we observe that the reviewed papers (2017 to 2023) are mostly interested in the NSL-KDD dataset because of its improved features and balance over the older KDD Cup 99 dataset. Therefore, it can be inferred that the methods presented specifically target DoS attacks on IoT devices. Moreover, some studies utilize datasets collected from real-world environments. We note that the KDD Cup 99 dataset is still used. However, this dataset may not capture new attack patterns. In addition, we remark that most of the datasets focus on DoS, R2L, and U2R attacks, missing newer attack types such as Common Vulnerabilities and Exposures (CVE), zero-day vulnerabilities, and attacks that target hardware.

From this review, the commonly used algorithms include SVM, RF, ANN, DNN, and CNN, with results averaging between 95% and 98% accuracy and precision. To be more precise, SVM, ANN and CNN focus on intrusion classification processes and achieve significant precision and accuracy (between 98% and 100%). However, each of these algorithms has its disadvantages. ANN, for example, has two major problems: less accurate detection for rare attacks like U2R or R2L, which are less common, and identifying patterns within large-scale datasets. Similarly, SVM can be less effective when dealing with large-scale datasets due to its high computational cost and memory usage. CNNs, while powerful in pattern recognition, can be computationally expensive and require a large amount of labelled data to achieve high accuracy.

6. Conclusion and perspectives

This paper reviews recent research on IoT intrusion detection, analysing advancements in ML methods from 2017 to 2023. It proposes a taxonomy based on ML algorithms, feature selection methods, datasets, and evaluation metrics. While progress has been made, detection rates need improvement, particularly for challenging attack types resembling normal traffic. Moreover, reducing false alarms and processing time remains essential. There is still work to be done to improve IoT IDS based on ML techniques.

In terms of future work, In the long term, we aim to update the NSL-KDD dataset with the recent attack. Therefore, we can use this dataset to train an ML model for the industrial IoT use case. Moreover, we plan to improve the IDS architecture in order to meet these challenges effectively.

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